

compliance handling are the key activities that fall under this.

Having uniform cloud management, monitoring and operational activities across CSPs is also a big challenge, and that is why third party cloud management tools like CloudBolt or Densify are used.

CloudOps and its activities

CloudOps (cloud operations) is defined as the management, delivery and consumption of software in a computing environment where there is limited visibility into an app's underlying infrastructure. It is primarily concerned with the ongoing process of refining business solutions and methods to optimise the availability, flexibility and efficiency of cloud services such that all business operations can be executed without any hiccup.

The following are the four pillars that any CloudOps team can use for building a strong IT infrastructure to smoothly carry out all processes.

- a. **Abstraction:** This decouples management from the underlying infrastructure so that all cloud servers, instances, storage devices, network infrastructure and other technologies can be managed under a single head.
- b. **Provisioning:** Administrators can allow end users to allocate their own machines and monitor their utilisation. Provisioning allows applications to request more resources when needed and release them if not required. The process is efficient and automated.
- c. **Policy driven:** This involves the creation and enforcement of policies in the cloud, which give limits to the activities of applications and users.
- d. **Automation:** This includes provisioning, user management, security and API management. These days, artificial intelligence plays a big role in automation. The main functions of CloudOps are:

- Keeping cloud services and infrastructure up and running 24x7x365.
- Managing, enforcing and updating service level agreements at regular intervals.
- Strong management of compliance as well as configurations.
- Meeting performance and capacity demands to reduce wastage and optimise efficiency.
- Timely recovery and migrations for cloud optimisations.

AIOps and more for cloud operational excellence

AIOps or artificial intelligence for IT operations, a term coined by Gartner, is designed to handle Big Data, machine learning (ML) and analytics platforms for greater automation. AIOps has a high maturity model and is a long-term solution to enable predictive optimisation based on usage patterns and historical data. It uses an ML driven approach to create models for predictive learning and also suggest areas of improvisation based on data patterns. It enables cost optimisation of production and non-production environments separately with low risk. Tools like CloudHealth and Densify are based on AIOps.

The five pillars of a well architected framework for cloud operations are: performance, availability/reliability, operational excellence, cost optimisation, and security/compliance. Cloud operations are slowly moving from DevOps (which relies on configuration management, CI/CD and pipeline handling) to CloudOps (which manages cloud infrastructure provisioning and management activities). CloudOps is now focusing more on AIOps, which uses machine intelligence in analysing logs and metrics, and proactively sends out alerts whenever there are issues with respect to performance or budget forecasts.

For example, Amazon DevOps Guru is an ML driven service which is fully managed and can improve application performance and availability. It gets data from Amazon CloudWatch, AWS Config, AWS X-Ray logs, AWS CloudTrail feeds and uses it to identify anomalous behaviour such as latency issues, error rates, resource level constraints, service level disruptions like API call failure or any such event failures. This service has an integrated dashboard to visualise its operational health based on continuous analysis by metrics and event analysis.

Based on the integrated machine learning knowledge base, Amazon DevOps Guru also enables resource level planning, and handles preventive maintenance activity by proactively notifying when the defined threshold is reached for resources like auto-scaling or API call volume.

Amazon DevOps Guru can also be integrated with Amazon OpsCenter or simple notification service (SNS) to connect to a third-party incident management system like ServiceNow, so as to automatically log tickets based on how critical an issue is.

AIOps for cloud platforms offers the following benefits.

1. **Improved service quality and efficiency:** By introducing automated solutions like self-monitoring, self-healing of services to automatically predict and handle failures, self-adapting solutions based on rules or policies, the service quality and overall efficiency in activities like infrastructure provisioning, monitoring, alerts and workflow operations can be improved substantially.
2. **Improved productivity from integrated DevOps solutions:** By improving build/deployment activities with integrated machine learning solutions like code reviews and regression tests and by integrating ticketing solutions with DevOps to handle ticketing requests